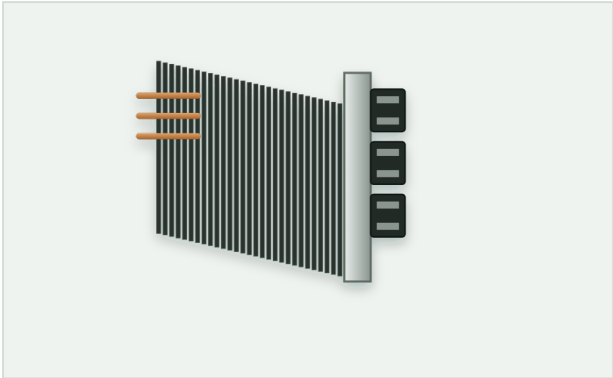
BOSARI
THERMAL MANAGEMENT

DATA SHEET

U-COOL® 1200/50

Heat pipe cooling system – Type 1200/50



- **1,200 W by natural convection**
- **6 copper “U” heat pipes, 29 couples of fins**
- **Weight approx. 19 kg**

THERMAL	
Total thermal power	**1200 W**
Type of cooling	**Natural air convection**
Rth (heatsink baseplate – ambient air)	**< 50 K/kW**

MECHANICAL	
Plate surface planarity	**0.05 mm**
Plate surface roughness	**1.2 µm**
Plate finish	**Chemical white alodine 1000 and further black painting on pipe side**
Fins finish	**Black epoxy painting**
Weight	**19 kg approx.**

ENVIRONMENTAL	
Operating position	**Vertical**
Tilting angle	**± 8°**
Ambient temperature range	**−25 °C ÷ +80 °C**
Shock and vibrations	**As per IEC 77 §16**

Description

Description

This type of Heat Pipe Cooler has been designed to meet the general specification prepared by one of our customers, regarding the cooling of an IGBT inverter for application on railway vehicles, mounted underfloor with fins oriented in a parallel plane to the moving direction.

Operation

The heat produced by IGBT modules, mounted on a base plate, is carried, by means of 6 “U” shaped heat pipes, to a radiating section composed by 29 couples of fins mechanically housed on the heat pipes. Fins in turn dissipate the heat into the ambient, through a natural air convection.

The particular shape of the cooler also provides a very high efficiency with the forced air, caused by the movement of the train (assisted ventilation).

Construction

The sample unit presented here basically consists of:

1. A base plate made of Aluminum alloy, machined on one face to house the lower leg of the heat pipes by means of brazings or mechanical clamping, and on the other face to allow the mounting of IGBT modules and Diodes by means of tapped threaded holes. On this same surface a groove for housing a gasket can be provided.
2. A series of 6 heat pipes made of Copper, housed and soldered to the base plate by means of a special soldering alloy. Thermocarrying fluid can be demineralised water or fluorinert. Pipes are inclined of 10° over the horizontal line to allow an operation mode gravity assisted.
3. A dissipating section consisting of 29 couples of fins made of Aluminum alloy, each having dimensions of 420x135x1 mm. Fins are separated in order to provide a better flow of the cooling air. Fins have six holes each to match the pipes size and are inserted on the pipes with a mechanical interference.
4. A special closing aluminum fin, of 3 mm thickness, is provided to ensure a stiffening of the pipes structure and an additional fixing point to the train structure.